



Lab Report

Only for course Teacher					
	Needs Improvement	Fair	Good	Excellent	Total Mark
Level of Content					
Content Development					
Spelling & Grammar					
Format					
Comments					

Semester: Fall 2025

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Student ID: 232-35-301

Batch: 41

Section: 41-H1

Course Code: SE313

Course Name: Software Quality Assurance & Testing Lab

Course Teacher Name: Muskan Ahmed

Designation: Lecturer

Submission Date: 15/12/2025



INTRODUCTION

For this lab, I worked with **Apache JMeter** it's basically a free and handy tool to test how well websites can handle traffic. What I did was create a complete test plan in JMeter to check the performance and behavior of some real-world apps. I set up things like thread groups to act like multiple users, HTTP samplers to send website requests, assertions to verify responses, listeners to track results, plus loop controllers and timers to control the test flow.

This time around, I picked **GitHub, Instagram, WhatsApp, and YouTube** as the apps to test. I wanted to see how these platforms respond when lots of users hit them at once, kind of like a real-world rush. Doing this helped me understand how they perform under pressure, and also gave me solid hands-on experience in setting up different JMeter tools to work together smoothly in one testing setup. Korba, I'll keep writing it this way straightforward and in everyday English.

TEST PLAN DESIGN

For this test plan, I set up four separate thread groups each one shows a different way JMeter can be used. While building it, I followed the instructions to add some key parts of JMeter and tweak certain settings. Here's what I included and why:

- **Thread Groups:** These basically tell JMeter how many fake users to simulate and how to run the test.
- **HTTP Samplers:** These are what send the actual web requests (like clicking a link or loading a page) to the server.
- **Listeners:** They collect the test results and show them so I can see what's happening.
- **Assertions:** I use these to check if the responses from the server are what I expected them to be.
- **Loop Controllers:** These let me repeat certain actions a set number of times during the test.
- **Timers:** I add these to put realistic pauses between requests, so it feels more like a real person is using the app.



Based on my student ID: (232-35-301), here's how I set things up in my JMeter test plan:

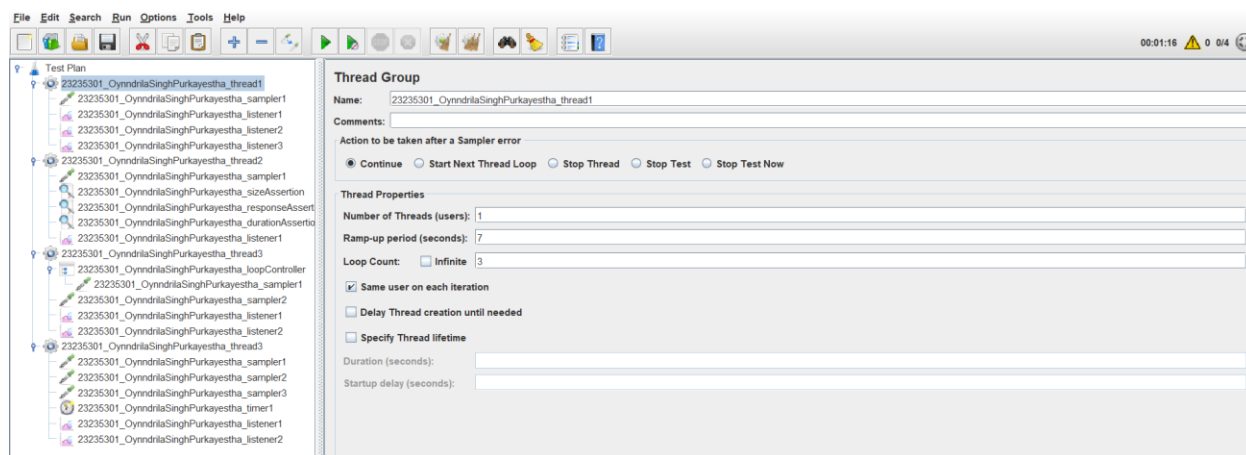
Since the last digit of my ID is 1, I used **1 user (thread)** and set the **loop controller count to 1** as well. The third last digit is 2, so I made the **loop count 3** (I just used a common rule for that one). I also added a **ramp-up period of 7 seconds**, so the user starts slowly. To make it feel real, I put in a **timer delay of 7000 ms** between requests. So basically, I matched all the settings to my ID as per the instructions.

Thread Group 1:

I name it "**23235301_OynndrilaSinghPurkayestha_thread1**". The main idea here was to send some basic requests to a website and see what happens when it gets a bit of user load. I also added a few different listeners so I could watch the results easily.

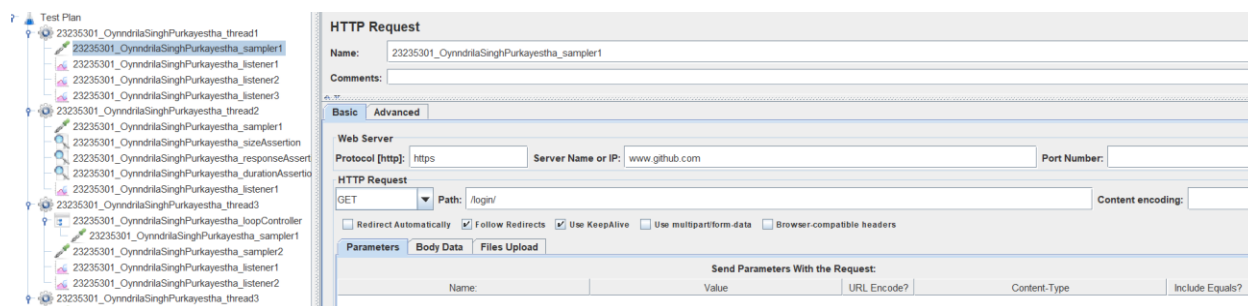
Here's how I set it up: I kept it to just **1 user**, made it start over **7 seconds**, and told it to repeat everything **3 times** in a loop. That's pretty much it for this thread group.

Thread Group 01 Screenshot:



In this thread group, I added an HTTP Sampler. I named it "23235301_OynndrilaSinghPurkayestha_sampler1". Its main job is to send a simple GET request to load the LinkedIn feed. For the web address, I used <https://www.github.com> and set the specific path to /login/

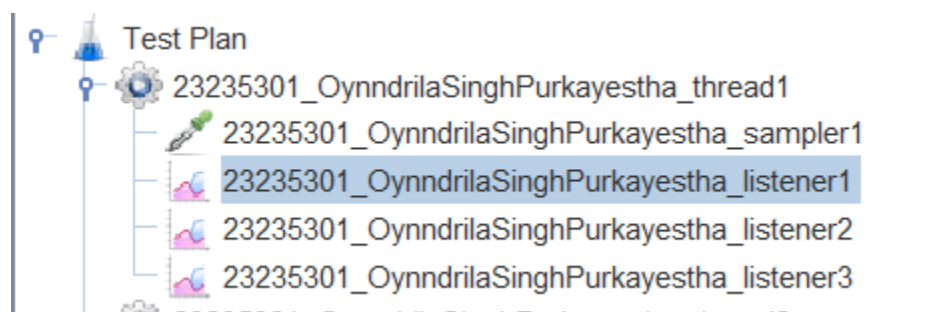
HTTP Sampler Screenshot:



Listeners:

23235301_OynndrilaSinghPurkayestha_listener1
23235301_OynndrilaSinghPurkayestha_listener2
23235301_OynndrilaSinghPurkayestha_listener3

Listeners screenshot:

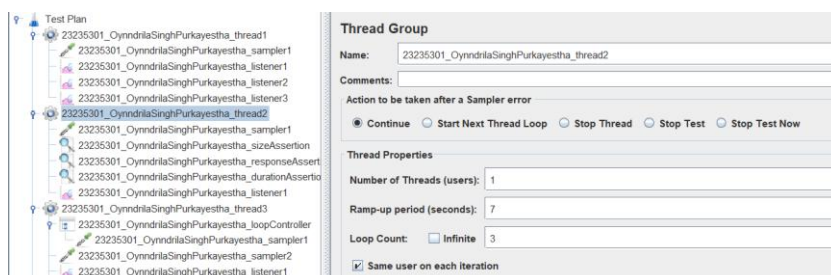


Thread Group 2:

I name it "**23235301_OynndrilaSinghPurkayestha_thread2**". The main idea here was to send some basic requests to a website and see what happens when it gets a bit of user load. I also added a few different listeners so I could watch the results easily.

Here's how I set it up: I kept it to just **1 user**, made it start over **7 seconds**, and told it to repeat everything **3 times** in a loop. That's pretty much it for this thread group.

Screenshot:

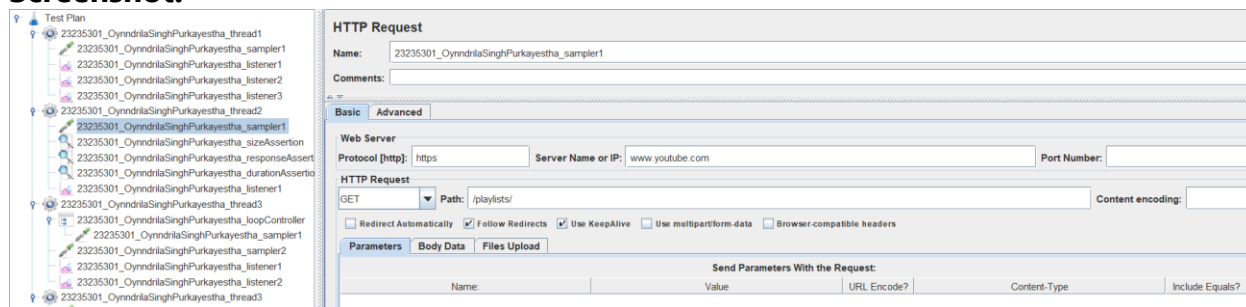




HTTP Sampler:

In this thread group, I added an HTTP Sampler. I named it "23235301_OynndrilaSinghPurkayestha_sampler2". Its main job is to send a simple GET request to load the LinkedIn feed. For the web address, I used www.youtube.com and set the specific path to /playlist/

Screenshot:



Assertions:

1. Response Assertion: I also put in a Response Assertion and called it "23235301_OynndrilaSinghPurkayestha_responseAssertion". In this test, its job is to check if what the server sends back actually has the text I'm looking for. To set it up, I told it to look at the Response Body, use the "equals" rule to match the text exactly, and I set the expected text to just say "youtube".

2. Duration Assertion: I also used a Duration Assertion and named it "23235301_OynndrilaSinghPurkayestha_durationAssertion". In my test plan, its purpose is to make sure the server doesn't take too long to reply. I set it up so that if the response time goes over 200,000 milliseconds, the test would mark it as a failure.

3. Size Assertion (Error): I also added a Size Assertion to my test plan. I named it "23235301_OynndrilaSinghPurkayestha_sizeAssertion". Its job was pretty simple just check that the amount of data the server sends back is exactly what I expect. For the setup, I told it to look for a response size of exactly 200 bytes.

Why Error?

The reason the test showed an error is that the actual size of the data coming back from the server wasn't 200 bytes exactly. I used the "Equals" setting in the Size Assertion, which means it only passes if the response size matches 200 bytes perfectly. What actually happened was the server sent back a much bigger response, so the sizes didn't match at all. That's why the Size Assertion failed even though the request itself worked and the page loaded just fine.



Screenshot of all assertion :

H1_Oyन्द्रilaSinghPurkayestha_23235301.jmx (C:\Users\hp\Downloads\H1_Oyन्द्रilaSinghPurkayestha_23235301.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

- 23235301_Oyन्द्रilaSinghPurkayestha_thread1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener3
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread2
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sizeAssertion
 - 23235301_Oyन्द्रilaSinghPurkayestha_responseAssert
 - 23235301_Oyन्द्रilaSinghPurkayestha_durationAssert
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread3
 - 23235301_Oyन्द्रilaSinghPurkayestha_loopController
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler2
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread3
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler2
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler3
 - 23235301_Oyन्द्रilaSinghPurkayestha_timer1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2

Size Assertion

Name: 23235301_Oyन्द्रilaSinghPurkayestha_sizeAssertion

Comments:

Apply to:
☐ Main sample and sub-samples ☒ Main sample only ☐ Sub-samples only ☐ JMeter Variable Name to use

Response Size Field to Test
☒ Full Response ☐ Response Headers ☐ Response Body ☐ Response Code ☐ Response Message

Size to Assert
Size in bytes: 1000

Type of Comparison
☒ = ☐ != ☐ > ☐ < ☐ >= ☐ <=

H1_Oyन्द्रilaSinghPurkayestha_23235301.jmx (C:\Users\hp\Downloads\H1_Oyन्द्रilaSinghPurkayestha_23235301.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

- 23235301_Oyन्द्रilaSinghPurkayestha_thread1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener3
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread2
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sizeAssertion
 - 23235301_Oyन्द्रilaSinghPurkayestha_responseAssertion
 - 23235301_Oyन्द्रilaSinghPurkayestha_durationAssertion
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread3
 - 23235301_Oyन्द्रilaSinghPurkayestha_loopController
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler2
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread3
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler2
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler3
 - 23235301_Oyन्द्रilaSinghPurkayestha_timer1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2

Response Assertion

Name: 23235301_Oyन्द्रilaSinghPurkayestha_responseAssertion

Comments:

Apply to:
☐ Main sample and sub-samples ☒ Main sample only ☐ Sub-samples only ☐ JMeter Variable Name to use

Field to Test
☒ Text Response ☐ Response Code ☐ Response Message ☐ Response Headers
☐ Request Headers ☐ URL Sampled ☐ Document (text) ☐ Ignore Status
☐ Request Data

Pattern Matching Rules
☐ Contains ☐ Matches ☒ Equals ☐ Substring ☐ Not ☐ Or

Patterns to Test

Custom failure message

1 200

H1_Oyन्द्रilaSinghPurkayestha_23235301.jmx (C:\Users\hp\Downloads\H1_Oyन्द्रilaSinghPurkayestha_23235301.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

Test Plan

- 23235301_Oyन्द्रilaSinghPurkayestha_thread1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener3
 - 23235301_Oyन्द्रilaSinghPurkayestha_thread2
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler1
 - 23235301_Oyन्द्रilaSinghPurkayestha_sizeAssertion
 - 23235301_Oyन्द्रilaSinghPurkayestha_responseAssertion
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 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler2
 - 23235301_Oyन्द्रilaSinghPurkayestha_sampler3
 - 23235301_Oyन्द्रilaSinghPurkayestha_timer1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener1
 - 23235301_Oyन्द्रilaSinghPurkayestha_listener2

Duration Assertion

Name: 23235301_Oyन्द्रilaSinghPurkayestha_durationAssertion

Comments:

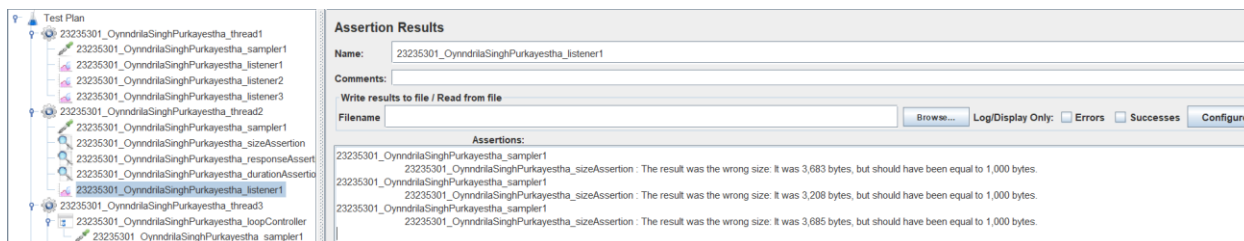
Apply to:
☐ Main sample and sub-samples ☒ Main sample only ☐ Sub-samples only

Duration to Assert
Duration in milliseconds: 200000



Listeners: I also added a listener called "23235301_Oyन्द्रilaSinghPurkayestha_listener1" to my test plan. Its job is to show me all the detailed information for each request and what the server sends back, like a report for every step in the test. I just used the default settings for it, so I didn't change anything special there.

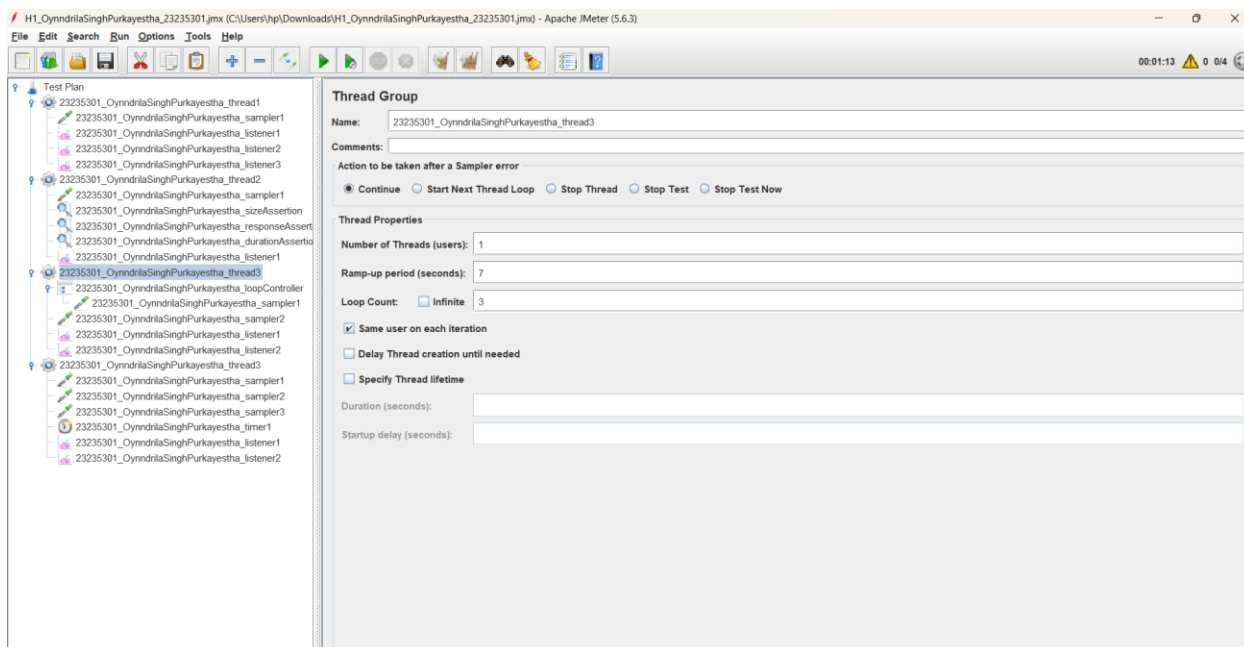
Screenshot:



Thread Group 3:

I name it "**23235301_Oyन्द्रilaSinghPurkayestha_thread3**". The main idea here was to send some basic requests to a website and see what happens when it gets a bit of user load. I also added a few different listeners so I could watch the results easily.

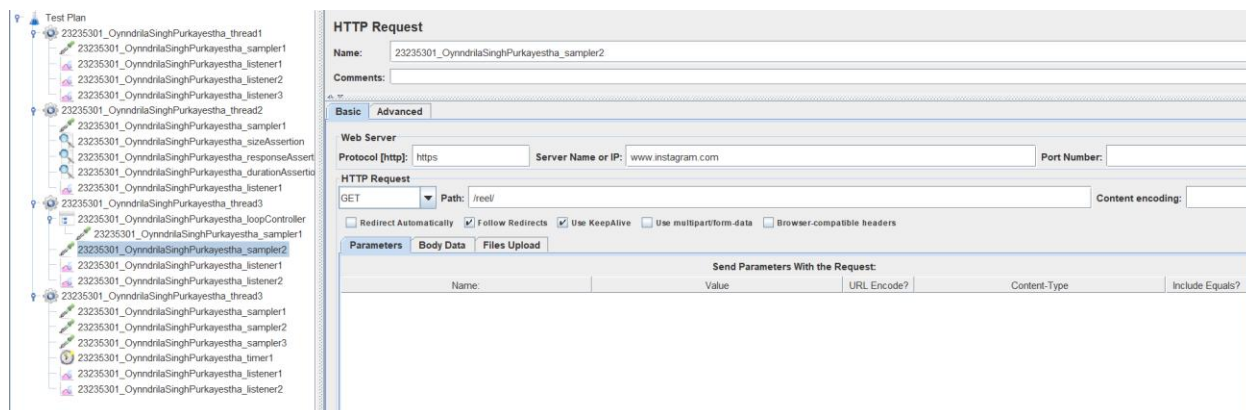
Here's how I set it up: I kept it to just **1 user**, made it start over **7 seconds**, and told it to repeat everything **3 times** in a loop. That's pretty much it for this thread group.





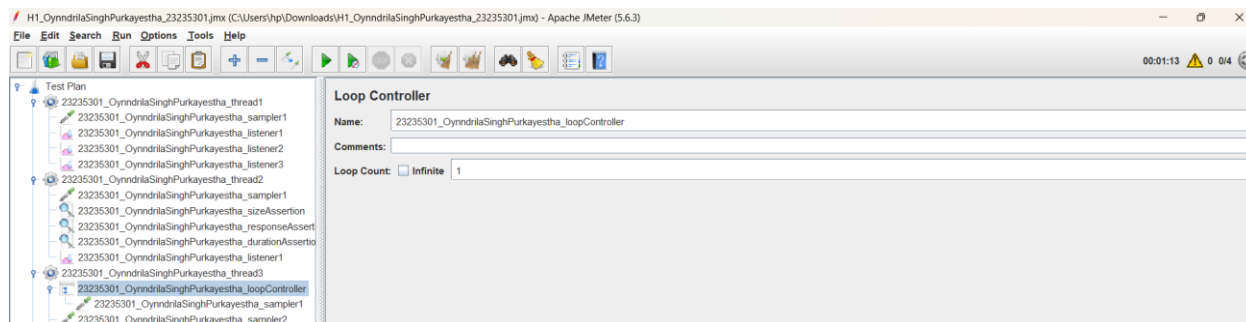
HTTP Sampler: In this thread group, I added two HTTP Sampler. I named it 23235301_OynndrilaSinghPurkayestha_sampler and "23235301_OynndrilaSinghPurkayestha_sampler2". Its main job is to send a simple GET request to load the LinkedIn feed. For the web address, I used www.instagram.com and set the specific path to /reel/

Screenshot:



Loop Controller: I added a Loop Controller to the test and named it "23235301_OynndrilaSinghPurkayestha_loopController". Its purpose was to make the sampler run more than once, based on how many loops I set. For the configuration, since the last digit of my student ID is 1, I set the loop count to 1.

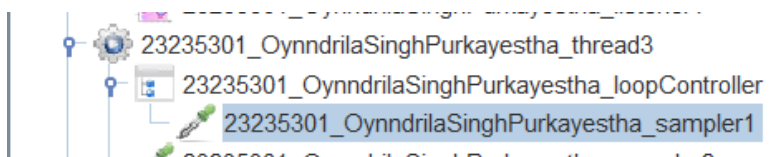
Screenshot:



HTTP Sampler inside Loop Controller

Inside the Loop Controller, I placed an HTTP Sampler named "23235301_OynndrilaSinghPurkayestha_sampler1". Its job in the test plan is to send a GET request to the Instagram homepage, and since it's inside the loop, it will do that repeatedly. I set the protocol to https for the connection.

Screenshot:

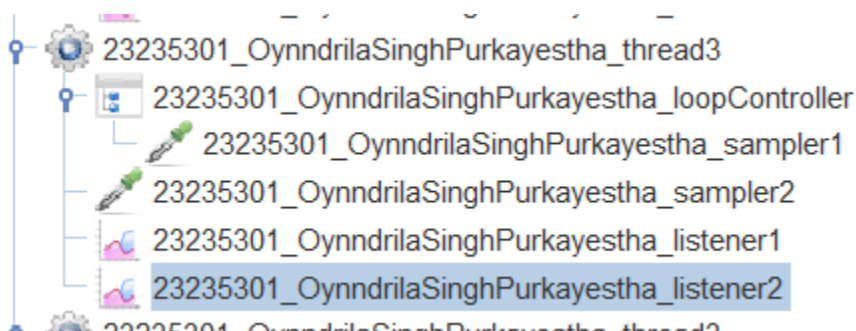




Listeners: For the listeners in my test, I added two different ones. The first is the View Results Tree, and I named it "23235301_OynndrilaSinghPurkayestha_listener1". Its job is to show all the detailed info for every single request and response, so I can check exactly what was sent and received. I kept its settings on default.

The second listener is the View Results in Table, which I called "23235301_OynndrilaSinghPurkayestha_listener2". This one helps me see performance trends in a simple table, like how long each response took and the overall throughput. I also used the default configuration for this one

Screenshot:

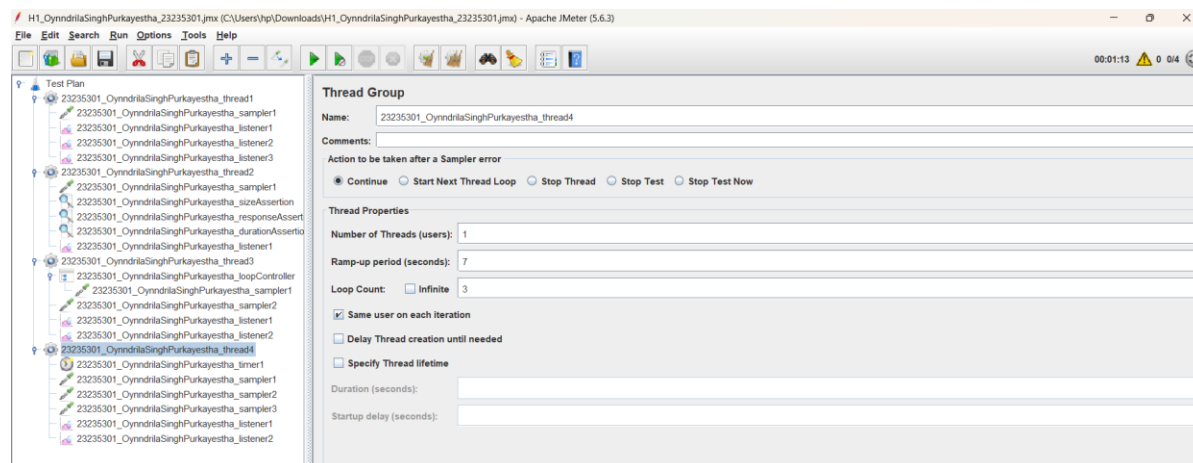


Thread Group 4:

I name it "23235301_OynndrilaSinghPurkayestha_thread4".

Here's how I set it up: I kept it to just **1 user**, made it start over **7 seconds**, and told it to repeat everything **3 times** in a loop. That's pretty much it for this thread group.

Screenshot:





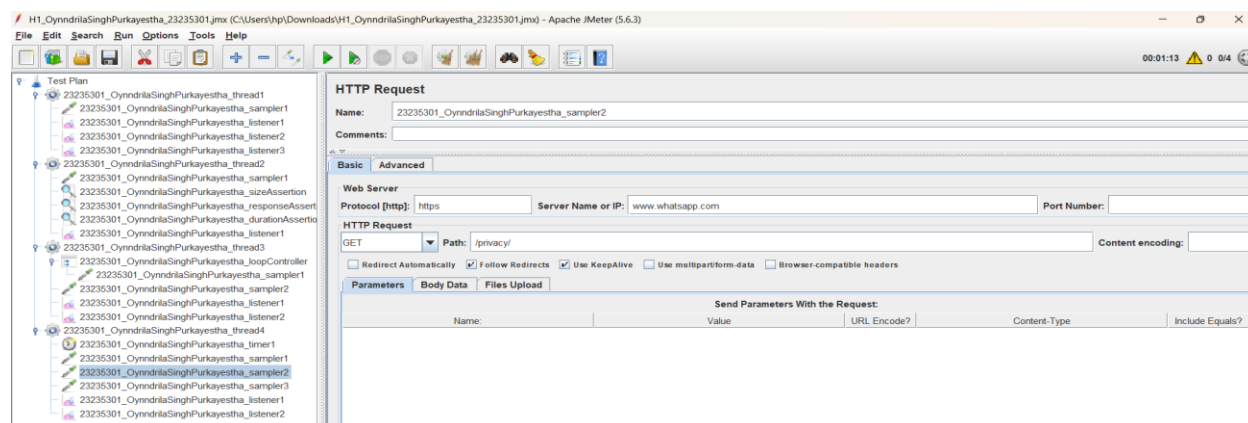
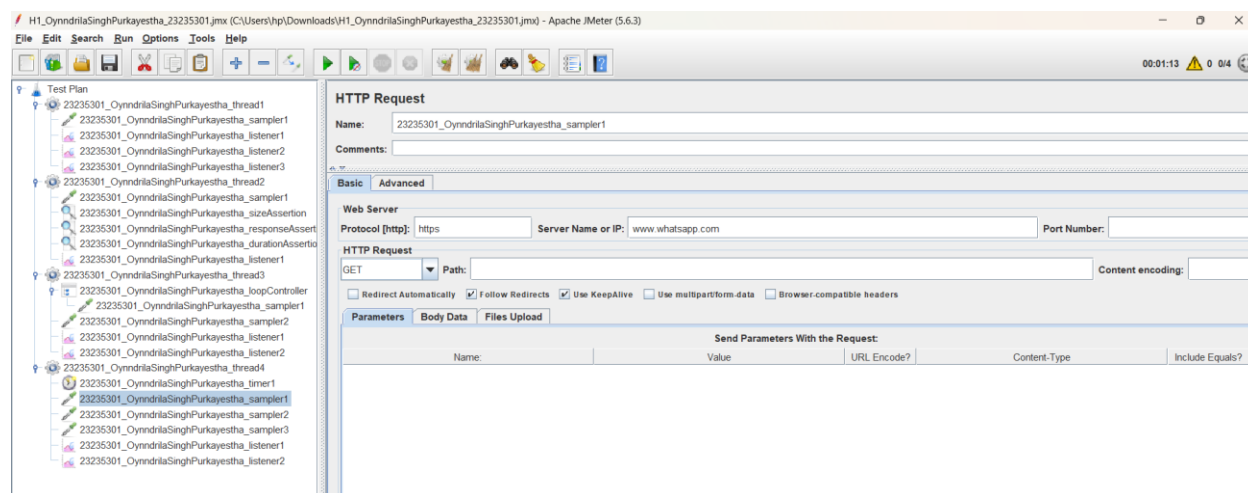
HTTP Samplers

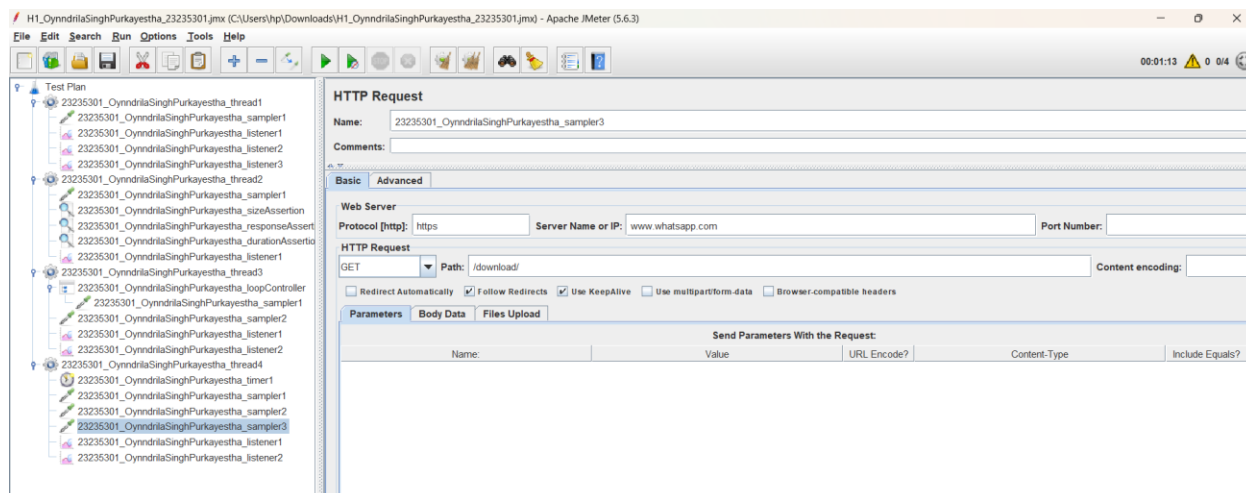
I set up three HTTP samplers in my test plan. The first one is named "23235301_OynndrilaSinghPurkayestha_sampler1". Its job is to send a simple GET request to the WhatsApp homepage. I set it to use https and pointed it to www.whatsapp.com without any extra path.

The second sampler is called "23235301_OynndrilaSinghPurkayestha_sampler2". This one also sends a GET request to WhatsApp, but this time it specifically targets the privacy page at the path "/privacy/". It uses https and the same server name.

The third sampler is "23235301_OynndrilaSinghPurkayestha_sampler3". This one sends another GET request, this time to the WhatsApp download page, which is at the path "/download". Like the others, it uses https and connects to www.whatsapp.com.

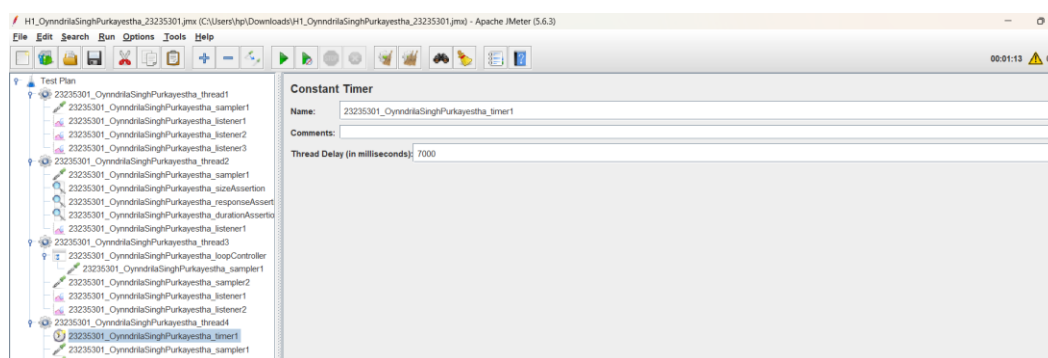
Screenshot:



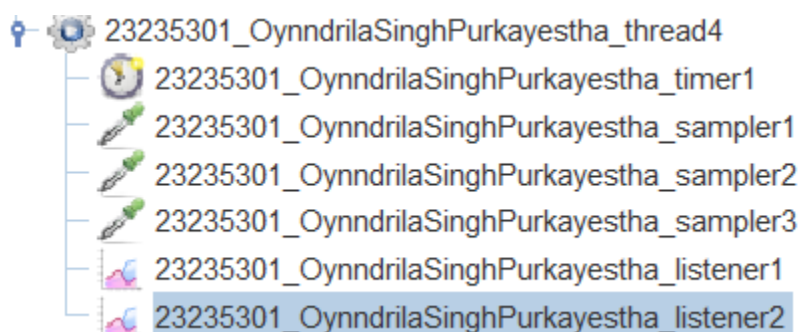


Timer: I added a Constant Timer to my test and called it "23235301_OynndrilaSinghPurkayestha_timer1". Its job is to create a fixed pause between one request and the next, so it feels more like a real person using the app instead of a machine sending requests non-stop. For the delay, I set it to 7000 milliseconds, which is 7 seconds, because the second last digit of my student ID is 0 and I followed the given rule.

Screenshot:



Listeners: 23235301_OynndrilaSinghPurkayestha_listener1 and 23235301_OynndrilaSinghPurkayestha_listener2



EXECUTION AND RESULTS

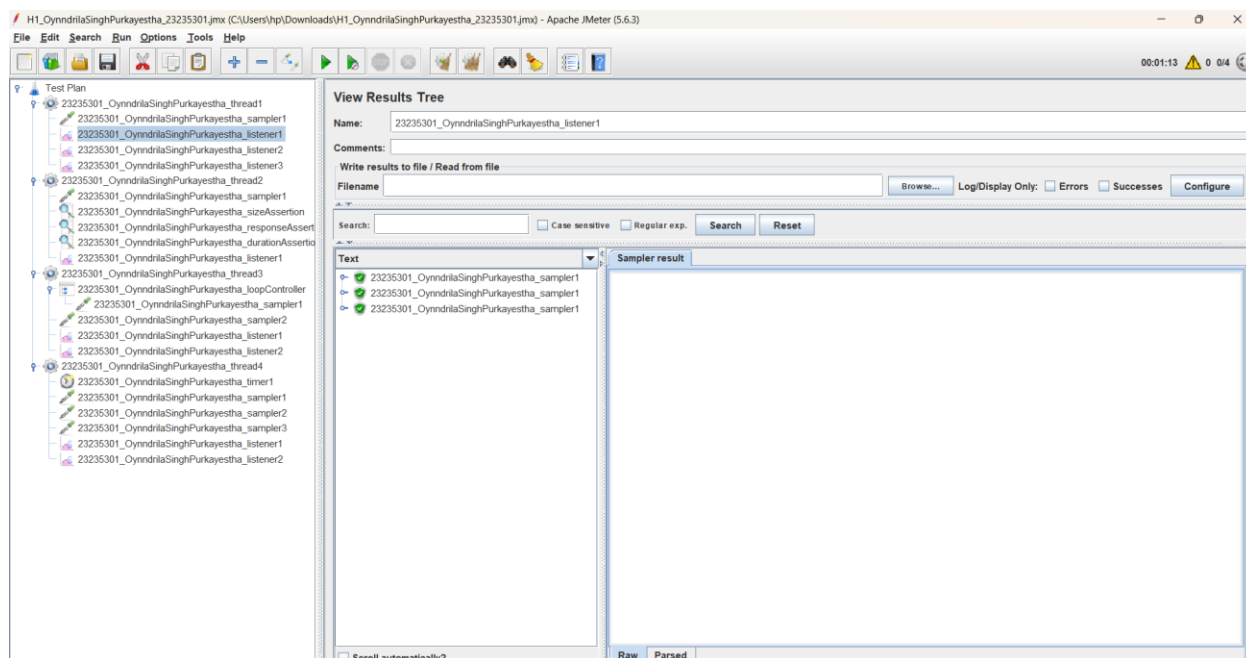
Thread Group 1:

Listener: View Results Tree

Name: 23235301_OynndrilaSinghPurkayestha_listener1

View Results Tree Analysis: For this test execution, the View Results Tree listener was used to observe and verify the execution status of the requests under Thread Group 1. From the results shown in the listener, all executed samplers are displayed with green success icons, which indicates that every request was successfully processed.

The sampler named 23235301_OynndrilaSinghPurkayestha_sampler1 was executed multiple times, and each execution completed successfully without any failures. No error responses, connection issues, or server-side problems were detected during this execution. This confirms that the requests sent from Thread Group 1 were handled correctly by the target server and produced valid responses.





Thread Group 1:

Listener: View Results in Table

Name: 23235301_OynndrilaSinghPurkayestha_listener2

View Results in Table Analysis: The View Results in Table listener shows that all three samples were executed successfully with no failures. The first request took longer time (1159 ms) due to initial connection setup, while the following requests were much faster (158 ms and 157 ms). Latency and connect time were also higher for the first request and reduced to near zero for subsequent requests, indicating connection reuse. The response size and sent bytes remained consistent for all samples, confirming stable and reliable system performance.

View Results in Table

Name:

23235301_OynndrilaSinghPurkayestha_listener2

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	03:46:05.347	23235301_Oynndril...	23235301_Oynndril...	1159	✓	47884	240	302	226
2	03:46:06.506	23235301_Oynndril...	23235301_Oynndril...	158	✓	47885	240	82	0
3	03:46:06.665	23235301_Oynndril...	23235301_Oynndril...	157	✓	47882	240	74	0

Thread Group 1:

Listener: Summary Report

Name: 23235301_OynndrilaSinghPurkayestha_listener3

Summary Report Analysis: The Summary Report shows that a total of three samples were executed with a 0% error rate, indicating all requests were successful. The average response time was 491 ms, with a minimum of 157 ms and a maximum of 1159 ms, showing some variation between initial and subsequent requests. The throughput was recorded at 2.0 requests per second, reflecting stable request handling by the server. Data transfer remained consistent, with an average response size of approximately 47.8 KB, confirming reliable and steady system performance during the test execution.

Ready system performance during the test execution.

Summary Report

Name:

23235301_OynndrilaSinghPurkayestha_listener3

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Label	# Samples	Average	Min	Max	Std. Dev.	Error %	Throughput	Received KB/sec	Sent KB/sec	Avg. Bytes
23235301_Oynnd...	3	491	157	1159	472.11	0.00%	2.0/sec	95.11	0.48	47883.7
TOTAL	3	491	157	1159	472.11	0.00%	2.0/sec	95.11	0.48	47883.7

Thread Group 2:



Listener: View Results Tree

Name: 23235301_OynndrilaSinghPurkayestha_listener1

Assertion Results Analysis: The Assertion Results listener indicates that the executed sampler failed the **Size Assertion** during multiple executions. Although the requests were processed successfully, the response size did not match the expected value of **1,000 bytes**. Instead, the actual response sizes were **3,683 bytes**, **3,655 bytes**, and **3,178 bytes**. This shows that while the server returned valid responses, the configured size assertion condition was not satisfied, resulting in assertion failures.

Assertion Results

Name: 23235301_OynndrilaSinghPurkayestha_listener1

Comments:

Write results to file / Read from file

Filename

Log/Display Only: ☐ Errors ☐ Successes

Assertions:

23235301_OynndrilaSinghPurkayestha_sampler1
23235301_OynndrilaSinghPurkayestha_sizeAssertion : The result was the wrong size: It was 3,683 bytes, but should have been equal to 1,000 bytes.

23235301_OynndrilaSinghPurkayestha_sampler1
23235301_OynndrilaSinghPurkayestha_sizeAssertion : The result was the wrong size: It was 3,655 bytes, but should have been equal to 1,000 bytes.

23235301_OynndrilaSinghPurkayestha_sampler1
23235301_OynndrilaSinghPurkayestha_sizeAssertion : The result was the wrong size: It was 3,178 bytes, but should have been equal to 1,000 bytes.

Thread Group 3:

Listener: View Results Tree

Name: 23235301_OynndrilaSinghPurkayestha_listener1

View Results Tree Analysis: The View Results Tree listener shows that all executed samplers were completed successfully, as indicated by the green success icons. Both sampler1 and sampler2 were executed multiple times, and no errors or failures were observed during the test run. This confirms that all requests sent to the server were processed correctly and returned valid responses without any connection or server-side issues.

View Results Tree

Name: 23235301_OynndrilaSinghPurkayestha_listener1

Comments:

Write results to file / Read from file

Filename

Log/Display Only: ☐ Errors ☐ Successes

Search: ☐ Case sensitive ☐ Regular exp.

Text

23235301_OynndrilaSinghPurkayestha_sampler1

23235301_OynndrilaSinghPurkayestha_sampler2

23235301_OynndrilaSinghPurkayestha_sampler1

23235301_OynndrilaSinghPurkayestha_sampler2

23235301_OynndrilaSinghPurkayestha_sampler1

23235301_OynndrilaSinghPurkayestha_sampler2

Sampler result



Thread Group 3:

Listener: View Results in Table

Name: 23235301_OynndrilaSinghPurkayestha_listener2

View Results in Table Analysis: The View Results in Table listener shows that all six samples were executed successfully with no errors. The response times ranged between 489 ms and 699 ms, indicating stable performance across requests. Latency values remained within a reasonable range, showing consistent server response behavior. The response size varied between requests, but all requests were processed correctly, and the sent bytes stayed nearly constant. Overall, the results indicate reliable execution and stable performance during the test run.

View Results in Table

Name:

23235301_OynndrilaSinghPurkayestha_listener2

Comments:

Write results to file / Read from file

Filename

Browse...

Log/Display Only: ☐ Errors ☐ Successes

Configure

Sample #	Start Time	Thread Name	Label	Sample Time(ms)	Status	Bytes	Sent Bytes	Latency	Connect Time(ms)
1	03:46:05.347	23235301_Oynndrila	23235301_Oynndrila	699	✓	774262	119	361	10
2	03:46:06.046	23235301_Oynndrila	23235301_Oynndrila	538	✓	863931	124	259	
3	03:46:06.584	23235301_Oynndrila	23235301_Oynndrila	524	✓	361727	119	282	
4	03:46:07.108	23235301_Oynndrila	23235301_Oynndrila	489	✓	865776	124	273	
5	03:46:07.597	23235301_Oynndrila	23235301_Oynndrila	525	✓	361303	119	291	
6	03:46:08.123	23235301_Oynndrila	23235301_Oynndrila	560	✓	866119	124	254	

Thread Group 4:

Listener: View Results Tree

Name: 23235301_OynndrilaSinghPurkayestha_listener1

View Results Tree Analysis: The View Results Tree listener shows that all executed samplers were completed successfully, as indicated by the green success icons. Multiple samplers (sampler1, sampler2, and sampler3) were executed repeatedly, and no errors or failures were observed during the test run. This confirms that all requests were processed correctly by the server and returned valid responses without any connectivity or server-side issues.

View Results Tree	
Name: 23235301_OynndrilaSinghPurkayestha_listener1	
Comments:	
Write results to file / Read from file	
Filename	
Browse... Log/Display Only: ☐ Errors ☐ Successes Configure	
Search: Case sensitive Regular exp. Search Reset	
Text	Sampler result
23235301_OynndrilaSinghPurkayestha_sampler1	
23235301_OynndrilaSinghPurkayestha_sampler2	
23235301_OynndrilaSinghPurkayestha_sampler3	
23235301_OynndrilaSinghPurkayestha_sampler1	
23235301_OynndrilaSinghPurkayestha_sampler2	
23235301_OynndrilaSinghPurkayestha_sampler3	
23235301_OynndrilaSinghPurkayestha_sampler1	
23235301_OynndrilaSinghPurkayestha_sampler2	
23235301_OynndrilaSinghPurkayestha_sampler3	

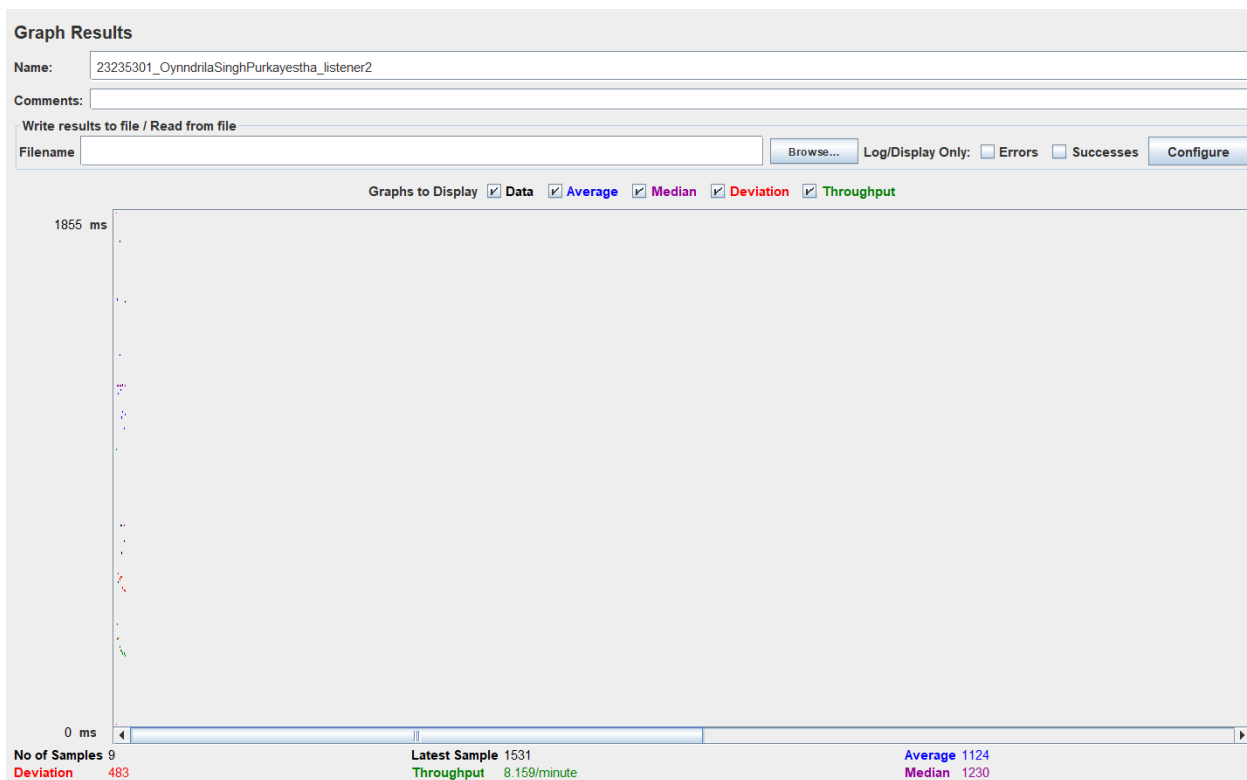


Thread Group 4:

Listener: Graph Results

Name: 23235301_OynndrilaSinghPurkayestha_listener2

Graph Results Analysis: The Graph Results listener shows the performance trend of the executed requests based on response time and throughput. A total of 9 samples were executed, with an average response time of 1124 ms and a median of 1230 ms, indicating moderate response variation. The standard deviation of 483 ms reflects some fluctuation in response times across requests. The throughput was recorded at approximately 8.15 requests per minute, showing steady request processing by the server. Overall, the graph indicates stable performance without any abnormal spikes or failures during the test execution.





DISCUSSION

Based on the analysis of all JMeter listeners (View Results Tree, View Results in Table, Summary Report, Assertion Results, and Graph Results), the overall test execution demonstrates stable and successful system behavior with some notable observations. Most of the requests were executed successfully, as confirmed by the View Results Tree and View Results in Table listeners, where all samplers showed green success indicators and no connection or server-side errors were detected. Performance metrics indicate that initial requests generally took more time due to connection setup, while subsequent requests were processed faster, showing efficient connection reuse and improved responsiveness.

The Summary Report further supports this by showing a **0% error rate**, acceptable average response times, and stable throughput, indicating that the system can handle the given load consistently. Graph Results illustrate moderate variation in response times, reflected by the standard deviation, but no abnormal spikes or failures were observed, suggesting overall performance stability under test conditions.

However, the Assertion Results reveal that the **Size Assertion failed**, as the actual response sizes did not match the expected value. This indicates a mismatch between test expectations and actual system behavior rather than a functional failure of the system itself. Therefore, while the application responses were valid and successfully returned, the assertion configuration needs adjustment to align with real response data.

In conclusion, the system performed reliably in terms of request handling and performance, but minor improvements are required in assertion configuration to ensure accurate validation. Overall, the test results confirm that the application is functionally stable and performs consistently under the executed test scenarios.

CONCLUSION

The JMeter test execution was completed successfully and demonstrated stable and reliable system performance. All requests were processed without any connection or server-side errors, as confirmed by the View Results Tree, View Results in Table, and Summary Report listeners. Performance metrics showed acceptable response times, consistent throughput, and efficient connection reuse after the initial requests.

Although the Size Assertion failed, this issue was due to a mismatch between the expected and actual response sizes rather than a functional problem in the system. Overall, the application responded correctly



and consistently under the tested load conditions. Therefore, it can be concluded that the system is functionally stable and performs well, with minor adjustments needed only in assertion configuration for more accurate validation.

REFERENCES

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Thank you